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Superseding
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MILITARY SPECIFICATION
SHELTER, ELECTRICAL EQUIPMENT
S-368()/MGC-9A

1. SCOPE

1.1 Scope.- This specification covers the transportable shelter facility designated as S-368()/MGC-9A which is part of the Central Office, Teletypewriter AN/MGC-9A. When part of the Central Office, it is capable of interconnecting or terminating 120 voice frequency teletypewriter or telephone circuits. (See 6.1 and 6.5)

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	Thread,Cotton
V-T-295	Thread,Nylon
QQ-A-200	Aluminum Alloy Bar,Rod,Shapes and Tube,Extruded,General Specification
QQ-A-225	Aluminum Alloy Bar,Rod,Wire or Special Shapes,Rolled Drawn or Cold Finish,General Specification for
QQ-A-250	Aluminum Alloy Plate and Sheet; General Specification for
QQ-S-571	Solder,Tin Alloy and Lead Alloy
QQ-S-698	Steel,Sheet and Strip,Low Carbon
QQ-S-766	Steel,Plate,Sheet and Strip - Corrosion Resisting
TT-E-516	Enamel,Lusterless,Quick-Drying,Styerated Alkyd Type

FSC 5815

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TT-E-527	Enamel,Alkyd,Lusterless
TT-E-529	Enamel,Alkyd,Semi-Gloss

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MIL-P-116	Preservation,Methods of
MIL-W-530	Webbing,Textile,Cotton,General Purpose, Natural or In Colors
MIL-T-3530	Thread and Twine,Mildew Resistant or Water Repellant Treated
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-4088	Webbing,Textile,Woven Nylon
MIL-W-6858	Welding,Resistance, Aluminum,Magnesium, Non-Hardening Steels or Alloys,Nickel Alloys,Heat-Resisting Alloys and Titanium,Spot and Seam
MIL-T-7928	Terminals,Lug and Splice,Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux,Soldering,Liquid (Rosen Base)
MIL-W-45205	Welding,Gas,Metal-Arc & Gas,Tungsten-Arc,Aluminum Alloys,Readily Weldable, for Structures,Excluding Armor
MIL-E-15090	Enamel,Equipment,Light-Gray Formula No. 111
MIL-W-27265	Webbing,Textile,Woven Nylon,Impregnated
MIL-E-46061	Enamel,Camouflage,Solar and Heat-Reflecting
MIL-S-55505	Shelter,Electrical Equipment(Containing Equipment) Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multilevel Continuous Sampling procedures & Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol shall be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the shelter facility for approval. (See 4.3)

3.2 Construction.-

3.2.1 Procurement model.- A procurement model of the S-368()/MGC-9A will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the Invitation for Bid, physical construction of the equipment shall conform to the model.

3.3.3 Drawings.- If supplemental Engineering Drawings are cited in bid/contract, those drawings shall be used to manufacture the parts they represent in the Procurement Model.

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3.2.3 Materials.-

3.2.3.1 Fabric and thread.- Fabric and thread shall conform to the following:

a. The color of fabric shall be Olive-Drab No.7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread. Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage, in order to provide for satisfactory fit of finished items both before and after they are immersed in water and then dried.

b. Thread shall be cotton per V-T-276 and be given mildew resistant treatment in accordance with MIL-T-3530, Class II, and nylon per V-T-295.

c. Webbing shall be cotton in accordance with MIL-W-530, Class 7; and nylon per MIL-W-4088 or Class R of MIL-W-27265.

3.2.3.2 Metals.- Metals shall be selected from the appropriate detailed specifications of the basic specification cited in 2.1 as follows:

a. Extruded aluminum shall be in accordance with QQ-A-200.

b. Structural aluminum shall be in accordance with QQ-A-225.

c. Sheet aluminum shall be in accordance with QQ-A-250.

d. Sheet and strip steel shall be in accordance with QQ-S-698.

e. Sheet,strip, and plate steel, corrosion resistant shall be in accordance with QQ-S-766.

3.3 Electrical installation.-

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3.3.1 Power system wiring and components.- Installation and wiring of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps,. lighting fixtures, distribution assembly system and wiring components constituting a three wire electrical power distribution system) shall be in accordance with the model. The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- There shall be point to point circuit continuity on all power-line circuit and the resistance of the continuous circuit shall not exceed 3 ohms when tested in accordance with 4.6.1.1.

NOTE: Test to meet the requirements of 3.3.1.1 in accordance with 4.6.1.1 shall be accomplished following completion of power system wiring and prior to mounting of AC power receptacles in the wireway.

3.3.1.2 Dielectric breakdown voltage.- The dielectric strength of insulation of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 10 seconds when tested in accordance with 4.6.1.2.

3.3.1.3 Insulation resistance.- The resistance of insulation between any two of the three circuit conductors and also between each of the two ungrounded power line conductors and facility ground terminal with the fluorescent light switch in the OFF position, shall not be less than 100 megohms when tested in accordance with 4.6.1.3. The insulation resistance between conductor number 1 and 4 or 1 and shelter ground shall not be less than 10 megohms when tested in accordance with 4.6.1.3 with the fluorescent light switch in the ON position.

3.3.1.4 Operation of power distribution panel.- With all operating components of the AN/MGC-9A installed (See 6.1), all components on all circuits of the distribution panels shall meet operational tests. (See 4.6.1.4 thru 4.6.1.10)

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3.3.2 Signal (audio) system wiring and components.- Installation and wiring of signal wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, switches, terminal strips, terminal blocks, and wiring devices) shall be in accordance with the model. The completed in-place signal (audio) system installation in accordance with 4.6.2 shall meet the following requirements:

3.3.2.1 Continuity and circuit resistance.- There shall be circuit continuity between both end terminations of all signal-line conductors and the resistance of the continuous circuit in accordance with 4.6.2 shall not exceed 5 ohms when tested in accordance with 4.6.2.1.

3.3.2.2 Dielectric breakdown voltage.- The dielectric strength of insulation of all signal wires, cables and components shall withstand a potential of 600 volts DC applied continuously for not less than 5 seconds when tested in accordance with 4.6.2.2.

3.3.2.3 Insulation resistance.- The resistance of insulation between each signal-line conductor and all other signal-line conductors connected to ground shall not be less than 100 megohms when tested in accordance with 4.6.2.3.

3.3.3 Cord assemblies.- The following cord assemblies shall be fabricated in accordance with the model and meet MIL-C-3885. (See 4.4)

a. One (1) each Intercom cord assembly.
b. One (1) each Supervisor's phone cord assembly.
c. Eight (8) each CY-425()/U jack box patch cords.
d. One (1) each Air conditioner power cable.

3.3.4 Suppression of electromagnetic interference.- The motors in the ventilating equipment and heaters used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings (See 4.4)

3.4 System operation.- With all operating components (See 6.1) of the AN/MGC-9A installed and connected in accordance with the model, and with the shelter grounded, the equipment shall be capable of functioning in accordance with the following. (See 4.6.3 and 4.6.4)

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3.4.1 Signal Circuit Preparation.- Operating components shall be capable of being prepared and adjusted for operation as described in paragraph 4.6.4.1.

3.4.2 Signal Circuit Test.- The switchboards and teletypewriters shall be capable of functioning as described in paragraph 4.6.4.2.

3.4.3 Phone No.1 and Intercom test.- With telephones arranged for LB operation, communication shall be possible between supervisor's phone, intercom, and externally connected phone. (See 4.6.4.3)

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain without any extra tieing, taping, etc. by military vehicle without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.7)

3.6 Rail transportation.- The shelter facility without any extra tieing, taping, etc. shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail and shall be sub ected to and withstand the test of paragraph 4.8 without sustaining any buckling, cracking delamination, or other permanent deformation.

3.7 Drops, flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. After all drop tests of 4.9.1 and 4.9.2 there shall be no buckling, cracking delamination or other permanent deformation.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.10. During and after the test there shall be no leakage into the facility or its walls, floor, roof, or doors.

3.9 Fording.- The shelter facility shall be capable of withstanding the test of 4.11. During and after the test there shall be no leakage into the facility or its walls, floor, roof, or doors.

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3.10 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.12. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no permanent distortion of the shelter mounting surface or insert head or threads. Insert threads shall be free of paint.

3.11 Cleaning.-

3.11.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.11.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.12 Finish.- The shelter facility shall be finished in accordance with the following:

3.12.1 Exterior surfaces.- The exterior surfaces of the S-368()/MGC-9A such as for covers of boxes, side flaps on covers and shrouds for air-conditioner and blower, will have the same finish as the exterior surfaces of the bare shelter (finish P513 per MIL-F-14072; then lusterless white enamel No. 37875 per Fed Spec TT-E-516, 0.75 to 1.0 mil dry film thickness; then final film of lusterless green enamel per MIL-E-46061 (dry film thickness 1.5 to 2.0 mils).

3.12.2 Exterior hardware.- Exterior hardware of the S-368()/MGC-9A such as for entrance boxes, latches, braces, underside of covers, etc., will be finished the same as exterior surfaces except less the white enamel undercoat. These components will be painted prior to assembly to the shelter and any touch up will be with the lusterless green enamel of the shelter..

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3.12.3 Interior.- Finish per MIL-F-14072, P513. Follow with an application as below.

a. Ceiling, semi-gloss white enamel per Spec TT-E-529, color chip #27875 of Federal Standard No. 595.

b. Floor, lusterless gray enamel per Spec TT-E-527, color chip #36118 of Federal Standard No. 595.

c. Door and door brace, as in 3.12.1 above, to match exterior.

d. Walls, semi-gloss light green enamel per TT-E-529, color chip #24533 of Federal Standard No. 595.

e. Bracketry, semi-gloss light gray enamel per MIL-E-15090, Class 2. If bracketry is ferrous material and is not aluminum, finish per P213 of MIL-F-14072 instead of P513.

3.13 Marking.-

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the model.

3.13.2 Nameplates.- The shelter facility shall be provided with a nameplate. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft loading data plate.- The shelter facility shall be provided with two data plates similar to the one on the model. One data plate shall be imprinted with the dimensions which locate the centers-of-gravity, the gross weight and the bearing pressure on the bottom of the skids of the shelter facility. The other data plate shall contain the above information for the shelter assemblage. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the above information shall be determined by the contractor using the preproduction sample.

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3.13.4 Interior and exterior marking.- All interior and exterior marking of the equipment shall be as depicted on the model.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Seal,vapor.- All joints and raw edges of the framing for the air conditioner opening, power and signal entrance boxes, and exhaust blower, shall be sealed to provide a vapor barrier against entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one-part sealant,silicone construction sealant SE-1204, as made by General Electric Corp., or equal.

3.16 Securing of parts.- Brackets, lugs, flanges, and other mounting arrangements shall retain components of the facility securely so that loosening wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws,threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the model. Thread forming screws shall not be used.

3.17 Soldering.-

3.17.1 Solder and flux.- Soldering of electrical connections shall be made with composition SN60, Type S,R, or RMA in accordance with Federal Specification QQ-S-571. When type S is used, liquid rosin flux shall conform to MIL-F-14256. No acid or acid salts shall be used.

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3.17.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have come loose.

3.17.3 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.18 Tropicalization of material.- Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated in this specification as follows:

3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.18.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use for materials such as canvas, webbing, cloth, etc. (See 6.2.e)

3.19 Welding.- Welded joints shall be as shown in the model and shall be such that grinding on the finished weld will be unnecessary. Inert gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B and MIL-W-45205 CL.B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely nautralized and removed. (See 4.4)

3.20 Wiring and cabling.- Wiring and cabling shall be in accordance with the model and shall be neat and sturdy.

3.20.1 Slack.- Wires and cables shall be as short as practical except that slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing at hinges parts.

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3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.20.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements of Specification MIL-T-7928. (See 4.4 and 4.13)

3.20.5 Identification of wiring.- Wiring shall be identified by colorcoding and wire pair marking, as indicated by the model.

3.21 Interchangeability.- Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or the equipment. (See 4.14) Individual items shall not be hand-picked for fit performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, and the like.

3.22 Workmanship.-The S-368()/MGC-9A shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.2 Construction
3.3 Electrical installation
3.10 Mounting inserts
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal,vapor
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization of material
3.19 Welding
3.20 Wiring and cabling
3.21 Interchangeability

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery).(See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.16)

4.3 Preproduction inspection.- This inspection will be performed by the contractor on the preproduction unit unless otherwise specified in the contract. (See 3.1) It shall consist of the the inspection specified in the subsidiary documents covering the items listed in para. 4.4, and the inspection for Group A, Group B, and Group C (See Tables I,II and III respectively). The preproduction inspection shall normally be performed in this order unless otherwise specified.

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a. Mounting inserts
b. Power wiring
c. Signal wiring
d. System operation
e. Vehicular transportation
f. Rail transportation
g. Flat drop
h. Rotational drops
i. Watertightness
j. Fording

The last tests performed shall be a repeat of the system operation tests. Remaining preproduction tests specified for Groups A,B and C inspections but not included in the above list shall be performed in any order satisfactory to the contracting officer.

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Suppression, EMI	3.3.4
Cord assemblies	3.3.3
Finish	3.12
Marking	3.13
Welding	3.19
Terminals	3.20.4

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specifications requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government- at its discretion - may perform all or part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6) Test equipment for Government verification shall be made available by the contractor.

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4.5.1 Group A inspection.- Each unit of equipment shall be inspected for conformance to all the inspection and test requirements of Table I of this specification. Equipment conforming to the above requirements shall be verified by the contractor, utilizing the procedures for multilevel continuous sampling plan (CSP-M) of MIL-STD-1235, unless otherwise specified.

Table I - Group A inspection

Inspection	Rqr Para	Insp Para	AOQL % Def Major	AOQL % Def Minor
Visual and Mechanical	3.22	4.15	1.5(AQL-1.0%)	5.0(AQO-4.0%)
Electrical				**
Power wiring	3.3.1	4.6.1		
Continuity	3.3.1.1	4.6.1.1		100%
Dielectric strength	3.3.1.2	4.6.1.2		Inspection
Insulation resistance	3.3.1.3	4.6.1.3		for
				the
Power distribution	3.3.1.4	4.6.1.4 thru 4.6.1.10		entire group of
Signal wiring	3.3.2	4.6.2		power and signal wiring
Continuity & circuit resistance	3.3.2.1	4.6.2.1		
Dielectric strength	3.3.2.2	4.6.2.2		
Insulation resistance	3.3.2.3	4.6.2.3		

** All defects are major

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4.5.2 Group B inspection.- This inspection, including sampling shall conform to Table II and to the procedures for multi-level continuous sampling plan (CSP-M) of Standard MIL-STD-1235, unless otherwise specified. Group B inspection shall be performed on units of production that have passed Group A inspection except for Mounting Insert and Interchangeability Inspections which are performed prior. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

Table II - Group B Inspection

Inspection	Rqr Para	Insp Para	AOQL %Def (see 4.5.2.1)
Mounting insert	3.10	4.12.2	5.0(AQL 4.0%)
Watertightness	3.8	4.10	3.0(AQL 2.5%)
Fording	3.9	4.11	3.0(AQL 2.5%)
Interchangeability	3.21	4.14	5.0(AQL 4.0%)

4.5.2.1 Group B sampling plans.- The Group B sampling plans for the AOQL's listed in Table II of this specification shall be selected from tables VII, IX and X of MIL-STD-1235 which are for f=½. The number of units in Production Interval shall be the entire contract quantity unless otherwise specified in the contract.

4.5.3 Group C inspection.- This inspection shall be as listed in Table III, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

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Table III - Group C Inspection

Inspection	Rqr Para	Insp Para
System operation	3.4	4.6.3 and 4.6.4
Vehicular transportation	3.5	4.7
Rail transportation	3.6	4.8
*Drops, flat and rotational	3.7	4.9
Solderless-terminal-lug-connection	3.20.4	4.13

* On shelter facilities selected for the drop tests the water-tightness test of 4.10 and the fording test of 4.11 shall be performed after the drop tests.

4.5.3.1 Sampling for inspection of drop tests.- One sample from the first 50 units or fraction thereof produced, and one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for inspection of system operation, vehicular and rail transportation.- The shelter facility selected for the first Group C drop tests shall be subjected to the system operations, vehicular and rail transportation tests prior to the performance of the drop tests. The order of testing shall be as specified in 4.3.

4.5.3.4 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure.

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Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken additional sample units shall be subject to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See Para. 6.4)

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all damage.

4.6 Electrical installation tests.- The tests of 4.6.1 thru 4.6.4.3 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115 volts AC, 60 cycles, single phase.

4.6.1 Power system and components installation tests.- Prior to conducting power wiring and components tests of 4.6.1.1, 4.6.1.2, 4.6.1.3, the following shall be accomplished:

a. Remove or disconnect the following electrical loads:

(1) All fluorescent,incandescent, and neon (power indicator and power distribution box) lamps.

(2) All fluorescent lamp starters.

(3) Blowers and heaters.

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so that they do not short to each other or to ground.

c. Disconnect all green wires from distribution box neutral bus.

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4.6.1.1 Continuity of conductors.- Continuity resistance which meets the requirement of 3.3.1.1 shall be performed on each powerline circuit in accordance with the following paragraphs. A calibrated low resistance ohmmeter or equivalent shall be used in performing the test.

a. Remove and extend all connected power receptacles from the wireway so that the outlet mounting brackets do not make electrical contact to the wireway. (See Figure 1)

b. Place the following circuit controls in the ON position:

(1) Fluorescent lamp switch

(2) Incandescent cold start switch

(3) Blackout bypass switch

(4) Both blower switches

(5) Main circuit breaker

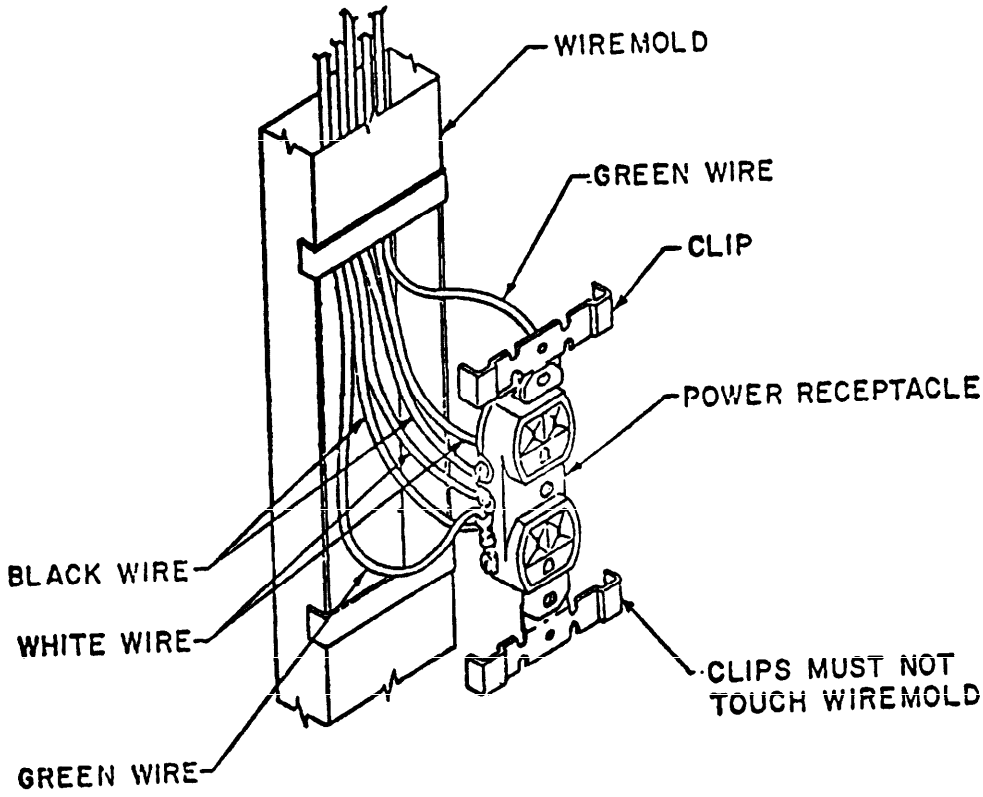
(6) All branch circuit breakers

c. Measure continuity from pin marked 1 of power connector IN located at power entrance box to the following: brass contact of each power receptacle and black wire termination at each fluorescent and incandescent lamp. Continuity measurements shall meet the requirement of 3.3.1.1.

d. Place all control listed under b above in the OFF position, except main circuit breaker.

e. Repeat step c above measuring continuity from pin 2 of power connector to the silver contacts of each power receptacle and white wire termination at each fluorescent and incandescent lamp.

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POWER RECEPTACLE REMOVED & EXTENDED FROM WIREMOLD

FIGURE NO. 1

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f. Place main circuit breaker in the OFF position.

g. Measure continuity from pin 4 (ground) of power connector to the following: each ground terminal of each power receptacle; and each ground terminal located at power entrance box, power distribution box, and video entrance box. Continuity measurements shall meet the requirements of 3.3.1.1.

h. Measure continuity between power connectors IN and OUT of the following pins: 1 to 1, 2 to 2, and 4 to 4. Measurements made shall meet the requirements of 3.3.1.1.

i. Replace and secure all power receptacles removed by step a above.

4.6.1.2 Dielectric breakdown voltage.- Dielectric strength tests which meets the requirements of 3.3.1.2 shall be performed on each power-line circuit in accordance with the following paragraphs. (Method 301 of MIL-STD-202, shall be employed using a nondestructive DC dielectric tester):

NOTE: Take preventive measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance tests.

a. Place all circuit controls listed under 4.6.1.1b in the ON position, except fluorescent lamp switch.

b. Test between pins 1 and 4 of power connector IN. Adjust tester for a value of not less than 500 volts DC output and maintain this potential for 10 seconds. The requirements of 3.3.1.2 shall be met.

c. Place fluorescent lamp switch in the ON position.

d. Repeat step b above with tester connected first across pins 1 and 2, and then across pins 2 and 4, of power connector IN.

e. Discharge each power-line conductor to ground following each test, to eliminate danger of shock.

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4.6.1.3 Insulation resistance.- Insulation resistance which meets the requirement of 3.3.1.3 shall be made on each power-line circuit in accordance with the following paragraphs. (Method 302 of MIL-STD-202 shall be employed using a megohmmeter or equivalent tester):

a. Place all circuit controls listed under 4.6.1.1b in the ON position, except blackout bypass switch; close shelter door securely.

b. Test between pins 1 and 4 of power connector IN. The output test voltage shall not be less than 100 volts DC nor more than 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has cease fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.1.3 shall be met.

c. Place blackout bypass switch to ON position and close shelter door securely.

d. Repeat step b. Requirements of 3.3.1.3 shall be met.

e. Repeat step b with tester connected first across pins 1 and 2, and then across pins 2 and 4, of power connector IN.

f. Discharge each power line conductor to ground following each test, to eliminate the danger of shock.

4.6.1.4 Operation of power distribution panel.- With shelter properly grounded using GROUND terminal in power and signal entrance box, connect power to the power in receptacle. The voltmeter on panel shall read nominal 115 VAC and neon lamp shall light. Place test lamps in each of the heater, equipment, and convenience receptacles.

4.6.1.5 Test of circuit breaker No.1.-

a. Place circuit breaker No.1, (Lights) in its ON position. Neon lamp shall light.

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b. Place BLACKOUT-BYPASS switch to BYPASS.

c. Place the two fluorescent lamp switches in their ON position. All fluorescent lamps shall light. Place incandescent cold start light switch in ON position.

d. Place the night-light switch (at supervisor's desk) in its ON position. The night-light shall light.

e. Place the BLACKOUT NORMAL switch in the BLACKOUT position.

f. With the shelter door open, all lights shall extinguish except the AC Power indicator lamp.

g. Close shelter door. All lamps shall light. Operate circuit breaker No.1 to assure lights are controlled by this breaker.

h. Leave fluorescent lamps on and turn off the night-light.

4.6.1.6 Test of circuit breaker No.2.-

a. Place circuit breaker No.2 (Equipment) in its ON position. Neon lamp shall light. Test lamps in the CV-425()/U and intercom positions shall light.

b. Place TT-98()/FG and TH-22()TG power switches located on the control panel between switchboard, to their ON positions.

c. Power indicator lamps shall light and test lamps in the TH-22()/TG and TT-98()/FG units shall light.

d. Operate circuit breaker No.2 to assure it controls the control panel. All test lamps in the above receptacles shall extinguish. Return all switches to their OFF position.

e. Turn off circuit breaker.

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4.6.1.7 Test of circuit breakers No.3 and No.7.-

a. Place circuit breaker No.3 (HEATER No.1) in its ON position. Lower neon lamp shall light.

b. Test lamp in No.1 heater receptacle shall light.

c. Operate breaker No.3 to assure it controls the heater receptacle.

d. Turn off breaker No.3. Test lamp in No.1 heater receptacle extinguishes.

e. Repeat steps a thru d for upper circuit breaker No.7 heater receptacle No.2 and upper neon lamp No.7.

4.6.1.8 Test of circuit breaker No.4.-

a. Switch on breaker No.4 neon indicator shall light.

b. Insert test lamp leads into air-conditioner receptacle on the outside of shelter.

c. Set air-conditioner thermostat fully clockwise or to lowest temperature (cooler).

d. Lamp in air conditioner receptacle shall light.

e. Operate circuit breaker No.2 to assure it controls the Air Conditioner.

f. Turn off circuit breaker No.4 and remove test lamp.

4.6.1.9 Test of circuit breaker No.5.

a. Place circuit breaker No.5 (Convenience 4) in its ON position. Neon lamp shall light.

b. Plug a test lamp into CONVENIENCE 4 (adjacent to signal wiremold on curbside wall). Test lamp shall light.

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c. Operate circuit breaker No.5 to assure it controls the outlet.

d. Repeat steps (b) and (c) for remaining CONVENIENCE 4 outlet on roadside wall.

e. Turn off circuit breaker.

4.6.1.10 Test of Circuit breaker No.6.-

a. Place circuit breaker No.6 (Blower) in its ON position. Neon lamp shall light.

b. Turn blower to its ON position. Blower shall exhaust shelter air.

c. Operate circuit breaker No.6 to assure it controls the blower.

d. Turn off blower and circuit breaker.

4.6.2 Signal (audio) system wiring and components installation tests.- Disconnect all signal lines from switchboard and other equipment.

4.6.2.1 Continuity and circuit resistance.- Continuity resistance which meets the requirements of 3.3.2.1 shall be performed on each audio line circuit in accordance with the following paragraphs. A low resistance continuity tester or equivalent shall be used. (An automatic tester may be used, subject to government approval).

a. Connect tester between 26 pair signal connector J1, pin No. 1A at signal entrance box and its corresponding tinned end at the SB-86 switchboard position. Continuity shall exist and meet the requirements of 3.3.2.1 (except pairs 25 and 26 which are terminated at PHONE 1 and INTERCOM jacks at supervisors desk).

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b. Connect tester between 26 pair signal connector J1, pin No. 1A and its corresponding binding post 1A at the signal entrance box (except pairs 25 and 26 as noted above). Continuity shall exist and meet the requirements of 3.3.2.1.

c. Repeat steps 4.6.2.1a and b for the remaining conductors of all 26 pair signal connectors.

d. Connect tester between each PHONE 1, INTERCOM, and spares binding post and its terminating end in the shelter.

e. Leave all signal lines disconnected to facilitate implementing paragraph 4.6.2.2.

4.6.2.2 Dielectric breakdown voltage.- Employing method 301 of MIL-STD-202 using a nondestructive DC dielectric tester (an automatic tester may be used, subject to government approval), dielectric strength tests which meet the requirements of 3.3.2.2 shall be performed on each audio-line circuit in accordance with the following paragraphs.

a. Using appropriate test jigs or test fixtures, connect the following to shelter facility ground terminal:

(1) All contacts, 1A thru 26B, of all five signal connectors.

(2) All binding posts in the signal and power entrance boxes.

b. Connect tester between shelter facility ground terminal and conductor 1A of signal connector J1. Remove ground connection from contact 1A of signal connector, system 1.

c. Adjust tester to 600 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.2.2 shall be met. Re-establish ground connection(s) to the contact(s) or binding post(s) under test.

d. Repeat steps b and c above for each of the remaining conductors of the same signal connector; each time ungrounding the contact of signal connector correlating with the circuit being tested and reconnecting to shelter facility ground following completion of test.

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e. Repeat steps b thru d above for each of the remaining signal connectors.

f. Repeat steps b thru e substituting binding posts for the contacts of signal connectors 1 and 2 respectively.

g. Connect tester between shelter facility ground terminal and Phone A binding post at power entrance box. Remove ground connections from contacts 25A of signal connectors 1 and 2; binding posts 25A of systems 1 and 2; and phone A binding post at power entrance box.

h. Repeat step c above.

i. Repeat steps g and h above for the other Phone, intercom, and spares binding posts at power entrance box, and shelter facility ground; each time ungrounding all contacts and binding posts correlating with the circuit being tested and reconnecting to shelter facility ground terminal following completion of test.

4.6.2.3 Insulation resistance.- Employing method 302 of MIL-STD-202, use a DC megohmmeter or equivalent tester; (An automatic tester may be used, subject to government approval). Insulation resistance tests which meets the requirements of 3.3.2.3 shall be performed on each audio-line circuit in accordance with the following paragraphs.

a. Repeat steps 4.6.2.2 a thru i above, except 4.6.2.2c, which shall be replaced by step b below.

b. Adjust tester to a value of not less than 100 nor more than 500 volts DC output. Maintain this potential for one minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.2.3 shall be met. Re-establish ground connection(s) to the contact(s) or binding post(s) under test.

4.6.3 System operation, utility units.-

4.6.3.1 Test of intercom.-

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a. Place circuit breaker No.2 (equipment) to its ON position. Neon lamp shall light.

b. Insert intercom plug into INTERCOM receptacle.

c. Place switch of intercom to its ON position.

d. After a few seconds neon lamp on intercom set LS-147 shall light.

e. Place circuit breaker No.2 to its OFF position.

f. Neon lamp on intercom set LS-147 shall go out.

4.6.3.2 Test of Heaters.-

a. Place circuit breaker No.3 (HEATER) in its ON position. Neon lamp shall light.

b. Place heater No.1 switch in the FAN position. Heater shall blow air.

c. Set thermostat to maximum temperature range.

d. Place switch in the HEAT position. After a short time the heater shall blow warm air.

e. Lower thermostat setting until a "click" is heard. This is the thermostat cutout and may be verified by observing a current drop in the power panel ammeter. The fan shall continue to operate.

f. With heater fan blowing, operate circuit breaker No.3 to assure it controls heater No.1.

g. Turn off heater.

h. Repeat steps (a) through (g) for circuit breaker No.7 and heater No.2.

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4.6.3.3 Test of air conditioner.-

a. Place circuit breaker No.4 (air conditioner) in its ON position. Neon lamp shall light.

b. Set Air Conditioner Thermostat fully clockwise (cooler).

c. Turn the OFF-VENT-COOL switch to cool. If the ambient temperature is 70 degrees or higher, the compressor shall operate and after a short time the Air Conditioner shall blow cold air.

d. Turn the OFF-VENT-COOL switch to the VENT position. Air Conditioner shall blow air without cooling operation.

e. Operate circuit breaker No.4 to assure it controls the Air Conditioner.

f. Turn off the Air Conditioner and circuit breaker.

4.6.4 System operation, signal circuits.-

4.6.4.1 Signal circuit preparation.-

a. All TT-98()/FG teletypewriters shall have at least 1/4 roll of paper and inking ribbons installed, and 2 batteries BA-30 at the rear of operators pack of SB-86/TG.

b. Unlock the keyboard, platen locks and carriage locks on each teletypewriter.

c. The receive (black plug) patch cord of each teletypewriter shall be plugged into the REC jack of its associated TH-22()/TG.

d. The send (red plug) patch cord of each teletypewriter shall be plugged into one of the SEND jacks of its associated TH-22()/TG.

e. Position POWER and LIGHT switches on both teletypewriters to ON. Teletypewriters shall be stable (not running open).

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f. Turn NORM-REC-SEND switch on each TH-22()/TG to NORM.

g. Turn 4W-2W-TEL screwdriver switch on each TH-22()/TG to TEL.

h. Turn MODE ONE - MODE TWO switch on each TH-22()/TG to MODE ONE.

i. Turn VF-20 screwdriver switch on each TH-22()/TG to 20 position.

j. Using a multimeter, adjust the loop current for each teletypewriter as follows:

(1) Position ON-OFF switch on TH-22()/TG to OFF.

(2) Connect the multimeter (set to measure current) in series with the receive loop.

(3) Obtain access to the INT BAT LOOP CUR (1R12) adjustment on the bottom of the TH-22()/TG.

(4) Position the ON-OFF switch on the TH-22()/TG to ON.

(5) Adjust INT BAT LOOP CUR (1R12) for a multimeter indication of 21 millamperes.

(6) Remove power from TH-22/TG (Step (1)), disconnect multimeter from receive loop and restore power to TH-22()/TG.

k. Position POWER and LIGHT switches on both teletypewriters to OFF.

1. Turn VF-20 screwdriver switch on each TH-22()/TG to VF.

m. Turn all line selector switches (located behind the designation strips) on each SWITCHBOARD SIGNAL ASSEMBLY TA-207()/P to M (Magneto).

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n. Position N.A. switch to VIS on each TA-207()/P.

o. Position LAMPS switch on each TA-207()/P to OFF.

p. Position TALK BAT switch on both switchboards to ON.

q. Position CIV TRKS switches on each TA-207 to the OFF position.

r. Open top cover of either POWER supply PP-990()/G.

s. Position BATT switch to INT.

t. Position INT SWBD BATT Switch to HI.

u. Position and hold BATT Check switch to 1. Meter shall indicate 20 to 26.5 volts. If meter indication is greater than 26.5 volts, position INT SWBD BATT switch to LOW. If meter indication is less than 20 volts position INT SWBD BATT switch to HI.

v. Position and hold BATT check switch to 2. Meter shall indicate 20 to 30 volts.

w. Release BATT CHECK switch and close top cover of power supply.

x. Repeat steps (r) thru (w) for remaining PP-990()/G.

y. Connect an operators Handset H-91()/U to the receptacle on each switchboard.

z. Turn CB-LB-CBS screwdriver switch on all TA-312()/PT telephones to LB position.

aa. Turn EXT-INT switch to INT.

ab. Turn loudness control on all TA-312()/PT telephones to LOUD.

ac. Place all cord circuit switches of each SB-86()/P switchboard in their neutral (handle straight up) position.

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ad. Equipment preparation test. Place circuit breaker No.2 (Equipment) in its ON position. Neon lamp shall light. Assure all equipment is in its proper positions and connected to its respective receptacle.

ae. Place POWER switch of CV-425()/U in its ON position. Power indicator lamp shall light.

af. Operate circuit breaker No.2 to assure it controls the CV-425()/U.

ag. Repeat steps (b) and (c) for the remaining three CV-425()/U units.

ah. Place TT-98()/FG and TH-22()/TG power switches located on the control panel between switchboards, to their on Positions. Power indicator lamps shall light.

ai. Place ON-OFF switch of each TH-22()/TG in its ON position. Power indicator lamps shall light.

aj. Place the ON-OFF-MOTOR switch of each TT-98()/FG in its ON position. Teletypewriter motors shall run.

ak. Operate the switches on the control panel to assure they control the TH-22()/TG and TT-98()/FG units.

al. Operate circuit breaker No.2 to assure it controls the control panel, then return all switches to their OFF position.

am. Turn off circuit breaker.

4.6.4.2 Signal circuits Test.-

a. The TH-22()/TG Telegraph Terminal, TT-98()/FG Teletypewriter, TA-312()/PT telephone and SB-86()/P switchboard grouped nearest the door end of the shelter shall be considered position 1. The same equipment grouped at the front of the shelter shall be considered as position 2.

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b. Connect a pair of wires, 12 ft. long, to the 2W-4WR binding posts on TH-22()/TG, position 1.

c. Connect a second pair of wires, 18 ft. long, to the 2W-4WR binding posts on TH-22()/TG position 2.

d. Connect pair #1 to SB-86()/P Lower TA-207()/P Line 1(EL) binding posts located in Binding Post Signal Entrance Box A.

e. Connect pair #2 to SB-86()/P Upper TA-207()/P Line 1 (E59) binding posts located in Bidning Post Signal Entrance Box B.

f. Lift the handset of TA-312()/PT, position 1, and actuate ringer. Line signal #1 on lower TA-207 of SB-86()/P shall turn from black to white and the panel lamps shall light.

g. On SB-86()/P, position 1, place the cord circuit switch of cord circuit #1 to the TALK LIST position.

NOTE: For purposed of this procedure the cord circuits of both switchboards shall be considered as being numbered consecutively from left to right.

h. Place the answer cord plug (in line with the cord circuit switch just moved - top cord) into line jack #1 (directly below the line signal indicating white) Line signal turns from white to black and panel lamps extinguish.

i. Communication between switchboard operator and the operator of TA-312()/PT, position 1, shall be loud and clear.

j. Place the call cord plug of this cord circuit into line jack #1 on upper TA-207()/P of SB-86()/P.

k. Place the ringing circuit switch to RING FWD. TA-312()/PT, position 2, shall ring.

l. Lift handset of TA-312()/PT position 2. Communication between TA-312()/PT, position 2, switchboard operator, and TA-312()/PT position 1, shall be loud and clear.

m. Place cord circuit switch of cord circuit #1 in neutral (handle straight up) position.

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n. Actuate ringers of both TA-312()/PT telephones. Supervisory signals of the cord circuit being used shall turn from black to white.

o. Pull the plugs of both cords from the line jacks and return them to their plug seats. Supervisory signal shall turn from white to black.

p. Repeat steps (d) thru (o) using signal lines and cord circuits listed below:

NOTE: At any convenient point during the tests, position the N.A. switches on switchboard SB-86()/P, Position 1, to AUD. Actuation of ringers on the TA-312/PT telephones shall cause the switchboard alarms to ring.

SB-86()/P Lower TA-207()/P Position 1	SB-86()/P Upper TA-207()/P Position 1	Cord CKT
Line 2 (E2)	Line 2(E60)	2
Line 3 (E3)	Line 3(E61)	3
Line 4 (E4)	Line 4(E62)	4
Line 5 (E5)	Line 5(E63)	5
Line 6 (E6)	Line 6(E64)	6
Line 7 (E7)	Line 7(E91)	7
Line 8 (E8)	Line 8(E92)	8
Line 9 (E9)	Line 9(E93)	9
Line 10(E10)	Line 10(E94)	10
Line 11(E11)	Line 11(E95)	11
Line 12(E12)	Line 12(E96)	12
Line 13(E39)	Line 13(E97)	13
Line 14(E40)	Line 14(E98)	14
Line 15(E41)	Line 15(E99)	15
Line 16(E42)	Line 16(E100)	16
Line 17(E43)	Line 17(E101)	any
Line 18(E44)	Line 18(E102)	any
Line 19(E45)	Line 19(E105)	any
Line 20(E46)	Line 20(E106)	any
Line 21(E47)	Line 21(E107)	any
Line 22(E48)	Line 22(E108)	any

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Line 23 (E49)	Line 23 (E109) any
Line 24 (E50)	Line 24 (E110) any
Line 25 (E53)	Line 25 (E111) any
Line 26 (E54)	Line 26 (E112) any
Line 27 (E55)	Line 27 (E113) any
Line 28 (E56)	Line 28 (E114) any
Line 29 (E57)	Line 29 (E115) any
Line 30 (E58)	Line 30 (E116) any

q. Repeat steps (b) thru (p) using switchboard SB-86()/P, TA-312()/PT and the signal lines and cord circuits tabulated below for position 2.

SB-86()/P Lower TA-207()/P Position 2	SB-86()/P Upper TA-207()/P Position 2	Cord CKT
Line 1 (E13)	Line 1 (E71)	1
Line 2 (E14)	Line 2 (E72)	2
Line 3 (E15)	Line 3 (E73)	3
Line 4 (E16)	Line 4 (E74)	4
Line 5 (E17)	Line 5 (E75)	5
Line 6 (E18)	Line 6 (E76)	6
Line 7 (E19)	Line 7 (E79)	7
Line 8 (E20)	Line 8 (E80)	8
Line 9 (E21)	Line 9 (E81)	9
Line 10 (E22)	Line 10 (E82)	10
Line 11 (E23)	Line 11 (E83)	11
Line 12 (E24)	Line 12 (E84)	12
Line 13 (E27)	Line 13 (E85)	13
Line 14 (E28)	Line 14 (E86)	14
Line 15 (E29)	Line 15 (E87)	15
Line 16 (E30)	Line 16 (E88)	16
Line 17 (E31)	Line 17 (E89)	any
Line 18 (E32)	Line 18 (E90)	any
Line 19 (E33)	Line 19 (E117)	any
Line 20 (E34)	Line 20 (E118)	any
Line 21 (E35)	Line 21 (E119)	any
Line 22 (E36)	Line 22 (E120)	any
Line 23 (E37)	Line 23 (E121)	any

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Line 24	(E38)	Line 24	(E122)	any
Line 25	(E65)	Line 25	(E123)	any
Line 26	(E66)	Line 26	(E124)	any
Line 27	(E67)	Line 27	(E125)	any
Line 28	(E68)	Line 28	(E126)	any
Line 29	(E69)	Line 29	(E127)	any
Line 30	(E70)	Line 30	(E128)	any

r. Turn 4W-2W-TEL screwdriver switch on each TH-22()/TG to 2W.

s. Turn VF-20 screwdriver switch on each TH-22()/TG to 20.

t. Position POWER and LIGHT switches on both teletypewriters to ON.

U. Place TT-98()/FG and TH-22/TG power switches located on control panel between switchboards to their ON positions.

v. Verify that pair #1 (Step b)and pair #2 (Step c) are still connected between the TH-22()/TG's and binding posts SB-86()/P lower TA-207()/P line 30 and SB-86()/P upper TA-207()/P; line 30,of position 2.

w. Depress the RING spring-return rotary switch on TH-22()/TG position 1. Line signal #30 on lower TA-207()/P turns from black to white and the panel lamps light.

x. On SB-86()/P, position 2, place the cord circuit switch of cord circuit #16 to the TALK LIST position.

y. Place the answer cord plug (in line with the cord circuit switch just moved) into line jack #30. Line signal turns from white to black and panel lamps extinguish.

z. Place the call cord plug of cord circuit #16 into line jack #30 on upper TA-207()/P of SB-86()/P, position 2.

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aa. Place the ringing circuit switch to RING FWD. The TH-22()/TG, position 2, shall ring.

bb. On TT-98()/FG, position 1, depress the repeat and P keys simultaneously. The TT-98()/FG, position 2 shall copy without error the letter being transmitted by TT-98()/FG, position 1. After two (2) complete lines have been typed, release the keys on TT-98()/FG, position 1.

ac. Repeat step aa using TT-98()/FG, position 2, to transmit to TT-98()/FG, position 1.

ad. Turn the TP-TG switch on all four (4) CV-425()/U's to TP.

ae. Turn the 2W-4W switch on all four (4) CV-425()/U's to 2W.

af. Place the SENSITIVITY switch on each CV-425()/U to LOW.

ag. Using the patch cords provided, make the following connections:

(1) Connect CV-425()/U #3 LOOP (J15) to SB-86()/P #1 UPPER TA-207()/P line 18 (J18).

(2) Connect CV-425()/U #3 LINE (J17) to CV-425()/U #4 LINE (J21).

(3) Connect CV-425()/U #4 LOOP (J19) to SB-86()/P #2 UPPER TA-207()/P LINE 29 (J20).

(4) Connect SB-86()/P #2 UPPER TA-207()/P LINE 5 (J11) to CV-425()/U #2 LOOP (J10).

(5) Connect CV-425()/U #2 LINE (J12) to CV-425()/U #1 LINE(J8).

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(6) Connect CV-425()/U #1 LOOP (J6) to SB-86()/P #1 LOWER TA-207()/P LINE 24 (J9).

ah. Connect pair #1 (step b) to SB-86()/P UPPER TA-207()/P LINE 18 (E102) pair of binding posts located in BINDING POST SIGNAL ENTRANCE BOX B.

ai. Connect pair #2 (step c) to SB-86()/P LOWER TA-207 LINE 24 (E50) pair of binding posts located in BINDING POST SIGNAL ENTRANCE BOX A.

aj. Turn 4W-2W-TEL screwdriver switch on each TH-22()/TG to TEL.

ak. Turn VF-20 screwdriver switch on each TH-22()/TG to VF.

al. Lift the handset of TA-312()/PT, position 1, and actuate the ringer. Line signal #29 on upper TA-207()/P of switchboard SB-86()/P, position 2, shall turn from black to white and the panel lamps shall light.

am. On switchboard SB-86()/P, position 2, place the cord circuit switch of any cord circuit to TALK LIST.

an. Place the answer cord plug of this cord circuit into line jack #29 of upper TA-207()/P on switchboard SB-86()/P position 2. Line signal shall turn from white to black and panel lamps extinguish.

ao. Place the call plug of this circuit into line jack #5 of the upper TA-207()/P on Switchboard SB-86()/P, position 2.

ap. Place the ringing circuit switch to TING FWD. TA-312()/PT, position 2, shall ring.

aq. Lift the handset of TA-312/PT, position 2. Communication between switchboard operator and TA-312()/PT, position 1, shall be loud and clear.

ar. Restore both handsets to their cradles.

as. Place the cord circuit switch in the neutral (handle straight up) position.

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at. Actuate the ringers on both TA-312()/PT telephones. Supervisory lamps of cord circuit being used shall turn from black to white.

au. Pull the plugs of both cords from the line jacks and return them to their plug seats. White supervisory signals shall turn black.

4.6.4.3 Phone No.1 and intercom test.-

a. Place the wire chiefs' TA-312()/PT telephone cord into the TELEPHONE jack on the TELEPHONE-INTERCOM panel.

b. Connect pair #1 (step 4.6.4.2b) to PHONE binding posts located in POWER AND SIGNAL ENTRANCE BOX.

c. Lift the handset on TA-312()/PT, position 1, and actuate the ringer. Telephone at Wire Chief's table shall ring.

d. Lift the handset of the Wire Chief's telephone. Communication between phones shall be loud and clear.

e. Repeat steps a through d for each pair of binding posts located in the binding post signal entrance boxes identified as PHONE.

f. Repeat steps a through e for INTERCOM jack and binding posts.

g. Disconnect pairs #1 and #2 from binding posts and TH-22()/TG's and secure from operational test.

4.7 Vehicular transportation test.- The shelter facility shall be secured (and blocked to prevent shifting) to a Government furnished truck, 2½ ton, M-35 and transported over the specially prepared course described in 4.7.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph and 10 laps at 18 to 20 mph. A lap is defined as traversing the course in one direction. The shelter facility shall meet the requirements of paragraph 3.5 after completion of the test. (NOTE: The shelter facility shall be loaded to simulate the AN/MGC-9A using live and dummy loads) (See 6.2(b) for this test).

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4.7.1 Vehicular transportation course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored securely by any effective means. The surface on which the timbers are placed shall be hard and level.

4.8 Rail transport impact test.- The shelter facility, loaded as specified in paragraph 4.7, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a weighted gondola car. The test shall be performed on a level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown strapping shall be as specified in Figure 2 and shall meet the provisions of the test as specified in paragraph 4.8.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The weighted gondola car shall meet the provisions of the test as specified in paragraph 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the Governments technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. The test shall consist of four impacts. Two impacts shall be performed with the door end of the shelter facing the loaded gondola; and two impacts shall be performed with the door end facing in the opposite direction from the loaded gondola. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken - during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

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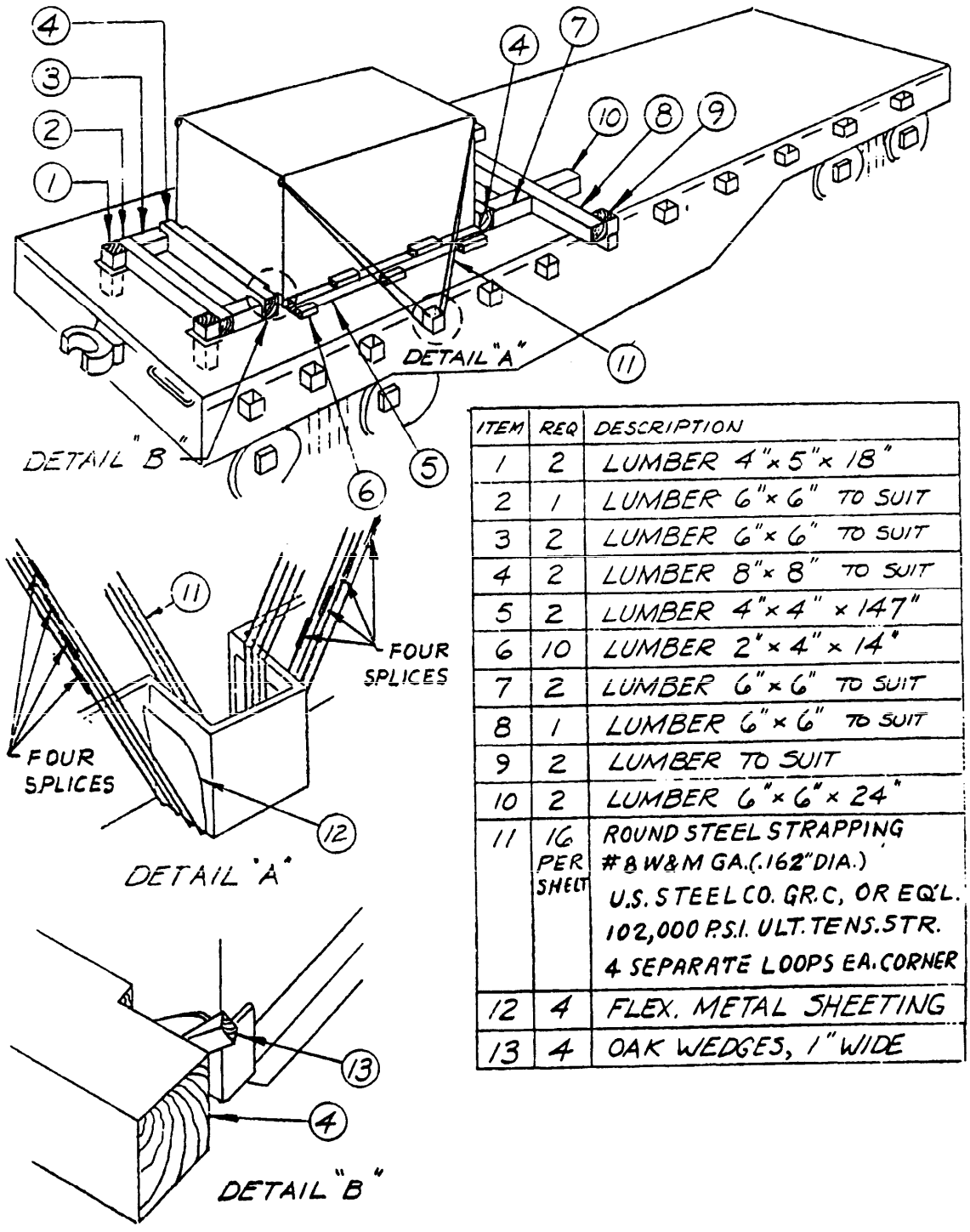
4.8.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 and 8 x 8 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The 8 x 8 inch blocking shall be notched to provide ¼ inch clearance around towing eye castings of the shelter; a wedge shall be installed into this clearance as shown in the figure referred to above, at each of the four corners of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All blocking around the shelter facility shall fit tightly against the shelter. All other blocking shall be firmly secured as shown in the figure. The tie down strapping shall be installed using a strapping machine (US Steel Corp. Model G.T. or equal) If tie down strapping loosens or breaks during any run, it shall be replaced before the next run with new wire and restrapped as above.

4.8.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x 5"x 18"
2	1	LUMBER 6"x 6" TO SUIT
3	2	LUMBER 6"x 6" TO SUIT
4	2	LUMBER 8"x 8" TO SUIT
5	2	LUMBER 4"x 4" x 147"
6	10	LUMBER 2"x 4" x 14"
7	2	LUMBER 6"x 6" TO SUIT
8	1	LUMBER 6"x 6" TO SUIT
9	2	LUMBER TO SUIT
10	2	LUMBER 6"x 6"x 24"
11	16 PER SHELT	ROUND STEEL STRAPPING #8 W&M GA.(.162"DIA.) U.S. STEEL CO. GR.C, OR EQ'L. 102,000 P.S.I. ULT. TENS. STR. 4 SEPARATE LOOPS EA. CORNER
12	4	FLEX. METAL SHEETING
13	4	OAK WEDGES, 1" WIDE

FIGURE 2 - SHELTER SECURED ON FLATCAR

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4.9 Drop tests.-

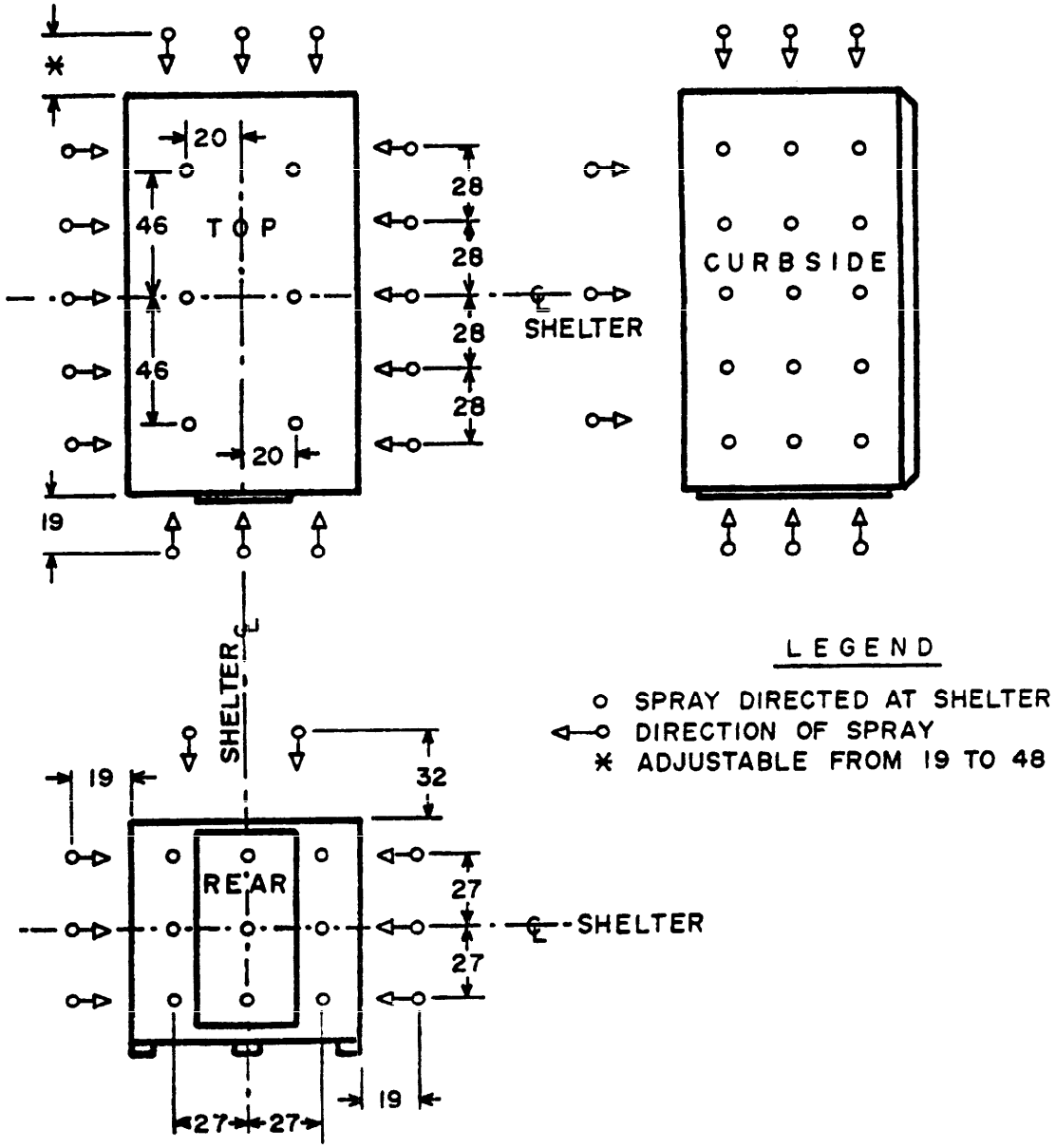
4.9.1 Flat.- The shelter facility loaded as in 4.7, shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. This test shall be performed once. Upon completion of the test, the shelter facility shall meet the requirements of 3.7.

4.9.2 Rotational drops.- The shelter facility loaded as in 4.7, shall have one skid (or the knuckles of the skid) resting on a nominal 4-inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and allowed to fall freely onto a hard substantial concrete surface. This impact test shall be performed four times, once for each bottom edge of the shelter facility. Upon completion of these drops, the shelter shall meet the requirements of 3.7.

4.10 Watertightness.- The shelter facility, without the use of any additional caulking, sealing, taping, etc., shall be subjected to the following two part test (NOTE: Perform tests with inside covers of entrance boxes removed):

a. Each of the five exposed surfaces of the shelter facility with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood,Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. For preproduction and Group C tests only, the air conditioner shall be installed. The row of nozzles in line with the air conditioner shall be 19 inches measured from the extreme vertical surface of the air conditioner. (See Figure 3 for nozzle location) All other testing shall be made with the blower blank panel used in place of the air conditioner. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. Each panel shall be sprayed for a period of twenty minutes. (If the watertightness test is to be followed by the fording test of 4.11, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 3
PAGE 45

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To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position and repeat (a) above.

4.11 Fording test.- The shelter facility shall be immersed in water to a depth of 21 inches (measured from the bottom of the skids) for a period of one hour and shall meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.12 Mounting insert tests.-

4.12.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with Table IV. After test the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, rack, etc.

4.12.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than 5, all of that size shall be tested) shall be randomly selected in each sample shelter facility. Each selected insert shall be tested in accordance with Table IV. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

Table IV.- Mounting insert loads

Insert Size	Preproduction Test Axial(lbs)	Torque(in-lbs)	Production Test Axial(lbs)	Torque (in-lbs)
10-32	600	23	300	23
1/4-20	1300	60	650	60
5/16-18	2000	100	1000	100

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4.13 Testing of solderless terminal-lug connections.- Connections made with solderless terminal lugs shall be tested to meet the requirements of 3.20.4.

4.14 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size, mounting dimensions of cabinets and cabinet drawers.

c. Size, mounting dimensions, and capacity of cooling and ventilization equipment.

4.15 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table V.

Table V.-Classification of Visual and mechanical defects

Classification	Defects
Major	1. Mounting insert missing or mislocated 2. Sealer improperly applied(holes,separations, or lack of adhesion) 3. Welds - cracked or porous 4. Parts, bracket, or equipment loose or missing 5. Drawers,doors,etc. inoperative 6. Mounting plates missing or mislocated 7. Aircraft loading data plate,instructions or nameplate omitted,incorrect or illegible. 8. Electrical receptacles,switches,connectors, wire ducts or other electrical components loose,improperly located or inoperative.

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Table V.-Classification of Visual and Mechanical defects (cont'd)

	9. Locking or holding devices missing or inoperative. 10. Burrs or sharp edges in wire ducts not removed. 11. Caps between sections of wire duct exceed 1/16".
Minor	1. Loose screws 2. Doors,slides,drawers not easily operated. 3. Finish-abrasions,scratches or chipped. 4. Locking or holding devices difficult to operate. 5. Burrs and sharp edges on brackets not removed.

4.16 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.17 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph 4.6.3	Systems operation
Paragraph 4.15	Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation,packaging, and packaging of the shelter facility shall be in accordance with applicable requirements of Specification MIL-S-55505.

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6. NOTES

6.1 Intended use.-

a. The Shelter,Electrical Equipment S-368()/MGC-9A is the shelter facility portion of the AN/MGC-9A and it is used to house the operational equipments listed below within the modified S-280()/G shelter.

2 each Switchboard,Telephone Manual SB-86()/P
2 each Switchboard,Signal Assembly TA-207()/P
2 each Terminal Telegraph TH-22()/TG
4 each Converter Telegraph-Telephone Signal CV-425()/U
2 each Teletypewriter TT-98()/FG
3 each Telephone Set TA-312()/PT
1 each Intercommunication Station LS-147()/FI

6.2 Ordering data.- Procurement documents should specify the following:

a. Title,number, and date of this specification and any amendment thereto.

b. Preproduction and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads for the communications security equipments (2 ea) and government furnished live loads (converter and telephone component) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-368()/MGC-9A on order.

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technicals means.

6.4 Group C inspection.- Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.4).

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-368()/MGC-9A becomes S-368A/MGC-9A. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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